

Project Development Plan

Project Development Plan

AI Assessment Tracking Assistant

1. Introduction

The **AI Assessment Tracking Assistant** is a smart academic management system developed to help students organize and monitor their academic assessments efficiently. In modern educational environments, students are required to manage multiple courses, assignments, quizzes, projects, and examinations simultaneously, making it difficult to remember deadlines, maintain schedules, and prioritize tasks effectively. This project is designed to solve these challenges by using artificial intelligence to provide automated academic tracking and smart reminder services. The assistant acts as a personal academic organizer that continuously keeps students informed about upcoming assessments and important deadlines.

➤ Purpose of the System

The purpose of the AI Assessment Tracking Assistant is to help students improve academic organization, productivity, and time management. The system is designed to track academic assessments and provide intelligent reminders before deadlines occur.

The assistant aims to:

- Help students avoid missing deadlines
- Reduce academic stress and last-minute panic
- Organize assignments and assessments in one place
- Improve task prioritization and planning
- Increase student productivity and consistency

➤ Main Features of the Assistant

Smart Reminder Notifications:-The system sends reminders at different times, such as:

- 7 days before the deadline
- 3 days before the deadline
- 1 day before the deadline
- On the submission day

Unified Assessment Calendar:-All assignments, quizzes, projects, and exams are displayed in one centralized academic calendar.

Progress Tracking Dashboard:-Students can monitor:

- Submitted tasks
- Pending tasks
- Overdue assessments

Task Prioritization:-The AI system can help students identify which assessments should be completed first based on urgency and importance.

Study Schedule Suggestions:-The assistant can recommend study or work schedules depending on upcoming deadlines and workload.

Task Completion Tracking:-Students can mark completed tasks and monitor productivity streaks over time.

2. Frontend Technologies

The frontend of the **AI Assessment Tracking Assistant** is developed using **React** and **Tailwind CSS**. These technologies are used to create the user interface that students interact with, such as dashboards, reminders, calendars, and task management pages.

React:- React is a JavaScript library used for building modern and interactive web applications. It was selected because it is fast, flexible, and allows developers to create reusable components. React helps make the system responsive and user-friendly.

React was chosen because:

- It provides fast performance
- It supports reusable components
- It makes the application interactive
- It works well with APIs and backend systems like Laravel

Tailwind CSS:- Tailwind CSS is a CSS framework used for designing the appearance of the application. It helps developers create clean and responsive designs quickly.

Tailwind CSS was selected because:

- It speeds up UI development
- It supports responsive design
- It provides a modern and organized interface
- It reduces the need for large CSS files.

3. Backend Technologies

The backend of the **AI Assessment Tracking Assistant** is developed using **Laravel**. Laravel is a powerful PHP framework used to manage server-side logic, databases, APIs, and user authentication.

Laravel was selected because it is secure, organized, and easy to maintain. It helps handle important system functions such as managing assessments, storing student data, sending reminders, and connecting the frontend with the database.

Laravel was chosen because:

- It provides strong security features
- It supports API development
- It simplifies database management
- It has clean and organized code structure
- It works well with React frontend applications

The backend also uses APIs to allow communication between the frontend and backend. APIs help the system send and receive data such as assignments, reminders, deadlines, and user information efficiently.

4. Database & Authentication

MySQL was selected because it is reliable, secure, and easy to integrate with Laravel. It also provides fast data storage and retrieval, which helps the system manage student information efficiently.

MySQL was chosen because:

- It securely stores user and assessment data
- It works well with Laravel
- It supports fast and efficient database operations
- It is widely used and easy to maintain

For authentication and security, Laravel's built-in authentication system is used. Authentication allows students to create accounts, log in securely, and protect their personal academic information.

The system includes security features such as:

- User login and registration
- Password encryption
- Secure authentication sessions
- Protection against unauthorized access

These security tools help ensure that student data and academic records remain safe and private. MySQL and Laravel authentication provide a secure and efficient database system for the AI Assessment Tracking Assistant.

5. AI & Notification Services

The **AI Assessment Tracking Assistant** uses the **OpenAI API** to provide intelligent features and smart assistance for students. The AI helps students manage assessments by giving reminders, organizing tasks, and supporting better academic planning.

The OpenAI API was selected because:

- It provides intelligent automation
- It improves task organization
- It helps generate smart reminders and suggestions
- It enhances the overall user experience

The system also uses **Email or notification services** to alert students about upcoming deadlines and important updates. Notifications are sent at different times before submission dates to help students avoid missing assessments.

6. Hosting & Free Tier Services

The **AI Assessment Tracking Assistant** uses **Vercel** and **Render** for hosting and deployment.

These platforms were selected because they provide free-tier hosting services. They also offer simple deployment processes, fast performance, and easy project management.

✓ **Vercel**

Vercel is used for frontend hosting because it works well with React applications and it works very well with React applications and provides fast and reliable deployment. Developers can connect their GitHub repository directly to Vercel, and whenever new changes are pushed to GitHub, the frontend application is automatically updated and deployed online.

Vercel also provides:

- Fast website performance ,Easy deployment process
- Automatic updates from GitHub
- Responsive hosting for modern frontend frameworks
- Free-tier services suitable for student projects

✓ **Render**

Render is used for hosting the backend part of the system, including Laravel APIs, server-side logic, and database communication. It manages important backend operations such as authentication, assessment management, reminder processing,

and API requests between the frontend and backend. Similar to Vercel, Render can also connect directly with GitHub repositories for automatic deployment and updates.

Render provides:-

- Backend server hosting ,API deployment support
- Automatic deployment from GitHub
- Secure server environment
- Free-tier hosting services

7. Workflow of the System

1. User Registration and Login

Students create an account and log into the system securely using authentication features provided by Laravel.

2. Adding Academic Information

Students add courses, assignments, quizzes, projects, and exam deadlines through the React frontend interface.

3. Data Storage

The Laravel backend processes the information and stores it securely in the MySQL database.

4. Assessment Tracking

The system continuously monitors upcoming deadlines and assessment schedules.

5. AI Reminder Processing

The AI service analyzes deadlines and prepares reminder notifications based on urgency and submission dates.

6. Notification Delivery

Students receive reminders through email or notification services at different times such as:

- 7 days before
- 3 days before
- 1 day before
- On the deadline day

7. Dashboard and Progress Monitoring

Students can view submitted, pending, and overdue assessments through the dashboard.

8. Task Completion Management

Users can mark assignments and assessments as completed to track their academic progress.

9. Hosting and Deployment

The React frontend is hosted on Vercel, while the Laravel backend is hosted on Render to keep the system accessible online.

8. Conclusion

The AI Assessment Tracking Assistant is a complete academic management system designed to help students efficiently organize and track their assessments, assignments, and deadlines. The system addresses common student challenges such as missed deadlines, poor organization, last-minute stress, and difficulty managing multiple courses by providing smart reminders, a unified dashboard, and AI-based task support. It is built using a modern and reliable technology stack where React and Tailwind CSS deliver a fast and responsive frontend, Laravel handles secure backend logic and APIs, and MySQL ensures safe and structured data storage. AI and notification services enhance the system by providing timely reminders and improved task management, while Vercel and Render offer simple and scalable free-tier hosting solutions. Overall, the integration of these technologies creates a user-friendly, efficient, and scalable platform that improves student productivity, time management, and academic performance.